

Headline part VII: the normal moment integral with arbitrary ratios

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Abstract

Paper-facing wrappers for the mixed-ratio general- d monomial milestone (units 177–181), stated on the Bochner box integral $M(N) = \int_{(0,1]^{m+1}} \prod_i x_i^{h_i} e^{-\beta N \prod_i x_i^{2k_i}} dx$: the exact weighted reduction, the dominated-coordinate transfer principle, the mixed-ratio asymptotic with the explicit Gamma/residue constant $\Gamma(\lambda)\beta^{-\lambda}/(|J|-1)! \cdot \prod_{i \in J} \frac{1}{2k_i} \prod_{i \notin J} \frac{1}{h_i+1-2k_i\lambda}$, and its form at the computed minimal ratio. The mirror `grammar_lean.tex` gains the labelled equation `eq:normal_moment_mixed_lean` inside `rem:normal_moment_lean` with a `\leanrefL` dot, and the pinned commit is bumped to `ef76bda`. Zero sorry/axiom.

1 The things formalised

```
theorem headline_normal_moment_weighted (d) (h k) (hk) ( N) :
  x in unitBox d, ( i, x i ^ h i) * exp (-( * N * i, x i ^ (2 * k i)))
= ( i, 1 / (2 * k i)) * (weightedBoxIntegral d
  (fun i => ((h i : ) + 1) / (2 * k i) - 1) (expKernel N)).toReal
theorem headline_dominated_coordinate ... (transfer lemma, unit 178)
theorem headline_normal_moment_mixed (m) (h k : Fin (m+1) → ) (hk) (l ) (hl) (h )
  (hmin : i, l ((h i : ) + 1) / (2 * k i))
  (hatt : i, ((h i : ) + 1) / (2 * k i) = 1) :
  (fun N => x in unitBox (m + 1),
    ( i, x i ^ h i) * exp (-( * N * i, x i ^ (2 * k i))))
~[atTop] fun N =>
  (Gamma l * ^ (-1) / (( i, if i = 1 then 1 else 0) - 1).factorial
    * i, if i = 1 then 1 / (2 * k i) else 1 / ((h i : ) + 1 - 2 * k i * 1))
    * N ^ (-1) * log N ^ (( i, if i = 1 then 1 else 0) - 1)
theorem headline_normal_moment_min (m) (h k) (hk) ( ) (h) : ... at minRatio h k
```

2 Paper-facing meaning

For the deterministic positive-orthant unit-cutoff normal moment integral, the paper’s zeta-function prediction is now formalised in full generality of exponent pattern and dimension: decay exponent equal to the minimal ratio, logarithmic degree one less than its multiplicity, and the Laurent coefficient $a_{-|J|}$ of eq. `a_minus_m_explicit` (at $b = 1$) times the Laplace–Tauberian factor $\Gamma(\lambda)/(|J|-1)!$. Excluded: cutoffs $b \neq 1$, interior parity assembly, variable amplitudes, stochastic phase terms, the full Taylor-tree expansion.

3 Lean notes

The wrappers spell out `multCount`, `monomialMixedConst` and `ratioExp` in the statement (sum of indicators, product of ifs), so the paper reader sees $|J|$ and the two residue products directly; the proofs are one-line `exacts` because those definitions unfold definitionally.